

COMPUTER NETWORKS

Week-15

Cloud Computing

- Availability of computing services - servers, storage, databases, networking, software, analytics and more - over the Internet (“the cloud”)
- Companies offering these computing services are called cloud providers and typically charge for cloud computing services based on usage

Characteristics

Shared Resources and Resource Management:

1. Cloud uses a shared pool of resources
2. Uses Internet technology to offer **scalable** and **elastic** services
3. The term “**elastic computing**” refers to the ability of **dynamically** and **on-demand** acquiring computing resources and supporting a variable workload

Characteristics

- 4. Resources are metered and users are charged accordingly
- 5. It is more cost-effective due to optimized resource utilization

Characteristics

- 6. Data is stored in the “cloud”, in certain cases closer to the site where it is used
- 7. Data appears to the users as if stored in a location-independent manner
- 8. The data storage strategy can increase reliability, as well as security, and can lower communication costs

Characteristics

8. The maintenance and security are efficiently operated by service providers

ADv

1. Resources, such as CPU cycles, storage, network bandwidth, are shared
2. Resources can be aggregated to support data-intensive applications
3. Data sharing facilitates collaborative activities
4. Many applications require multiple types of analysis of shared data sets
5. Manys decisions carried out by groups scattered around the globe
6. Eliminates the initial investment costs for a private computing infrastructure and the maintenance and operation costs

Adv

5. **Cost reduction:** concentration of resources creates the opportunity to pay as you go for computing
6. **Elasticity:** the ability to accommodate workloads with very large peak-to-average ratios
7. **User convenience:** virtualization allows users to operate in familiar environments rather than in unique environments with unique features

Types

1. **Public Cloud** - the infrastructure is made available to the general public or a large industry group and is owned by the organization selling cloud services.
2. **Private Cloud** - the infrastructure is operated solely for an organization.
1. **Hybrid Cloud** - composition of two or more Clouds (public, private, or community) as unique entities but bound by a standardised technology that enables data and application portability.
2. **Other types: e.g., Community/Federated Cloud** - the infrastructure is shared by several organizations and supports a community that has shared concerns.

Application Layer

Application Layer

- Today, most applications are networked applications
- Sets of interacting programs running on two or more hosts
- When a software developer writes a program, it is usually a program designed to talk to another program on a different host

Networked Applications

- Networked applications can use an infinite number of alternative interaction patterns
- → **application architectures** → patterns of interactions between networked applications on different machines

Evolution

- Stand alone programs (running on PCs and laptops) → Client/server architecture (PCs sharing processing chores with servers with WWW) → peer-to-peer computing (client providing server capabilities to another client : no need to have and manage servers) → distributed computing architecture (program on one machine calls programs on other machines - calling programs on other machines)

Cloud computing

- Think about ...
- Saving files on a laptop vs using Google Drive
- Installing software manually vs using Microsoft 365 online

Types of Cloud Services

- **IaaS**
- Infrastructure as a Service
- E.g. Virtual servers
- Storage
- Networking
- Business organizations now use computing infrastructure instead of buying expensive hardware
- E.g. Amazon Web Services EC2 (for virtual servers), Microsoft Azure Virtual Machines (support various OSs and scalable resources)

Types of Cloud Services

- **PaaS**
- Platform as a Service
- This is where developers use readymade environments to build applications
- E.g. Google App Engine (for developing and hosting web and mobile applications)
- It supports programming languages like Python, Java, Go, PHP etc.

Types of Cloud Services

- SaaS
- Software as a Service
- E.g. Zoom Communications
- Dropbox
- Salesforce (CRM platform)

Deployment Models of Cloud Computing

- Public Cloud
- IT infrastructure like servers, networking, storage resources are offered as virtual resources over internet
- Resources are fully managed and made accessible by the cloud provider
- Open to all users and organizations
- Less usage cost as users have to pay for what they use
- Less control over data → less storage cost as users pay only for what they use
- Highly scalable
- E.g. universities use public cloud

Public Cloud

- Public cloud security follows a shared responsibility model
- The public cloud provider is responsible for the security of the cloud
- They protect the infrastructure that runs all of the services offered in the cloud
- At the same time, the customer is responsible for security in the cloud
- The service determines the amount of configuration at customer end
- Some cloud computing services offer greater control and customizability but also increase the customer's security responsibility

Deployment Models of Cloud Computing

- **Private Cloud**
- A single organization controls and maintains the underlying infrastructure to deliver the IT resources
- Consider an organization with several departments, like finance and marketing, that need computing resources for their software applications
- In a private cloud setup, the organization purchases server hardware, maintains it in a central data center, and delivers the resources to the different departments over a network
- Individual teams may additionally invest in software infrastructure like operating systems or database software for their applications
- E.g. Banks using private cloud

Private Cloud

- Infrastructure is isolated for outside access
- Limited scalability as infrastructure is limited

Deployment Models of Cloud Computing

- Hybrid Cloud
- A company uses both the public and private cloud
- It is an IT infrastructure design that integrates a company's internal IT resources with third-party cloud provider infrastructure and service
- You can store your data and run your applications across multiple environments
- E.g. Companies combine both as hybrid environments

Hybrid Cloud

- Organizations typically adopt hybrid cloud strategies to overcome private cloud limitations
 - To continue using their existing on-premises data center
 - And ..
 - Access the public cloud as needed
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- Organizations which use hybrid approach have:
 - sensitive data
 - expensive IT infrastructure
 - Fluctuating workloads/requirements

Hybrid Cloud

- **Banks and financial institutions** may keep customer financial records, transaction data etc in private infrastructure
- These may use public cloud for mobile banking apps, data analytics, customer support systems etc
- **Universities** may store student records, exam data, internal administration data on private cloud.
- They may use public cloud for online learning resources, collaboration tools, research computing. E.g. MS Teams

Hybrid Cloud

- **Healthcare** organizations may use private cloud for patient medical records and internal hospital databases
- These may use public cloud for appointment portals, analytics etc.
- **Govt organizations** may use private cloud for storing national DBs, defense systems, citizens data
- Govt may use public cloud for their online services (tax services, passport services, smart city platforms), digital portals

Hybrid Cloud

- Think about **Retail and E-commerce** companies
- What may go on private cloud/systems?
- What may go on public cloud?

Hybrid Cloud

- Think about Retail and E-commerce companies
- What may go on private cloud/systems?
- Payment processing, inventory details/DBs
- What may go on public cloud?
- Online shopping websites
- Customer analytics
- Promotional campaigns

More Examples

- Netflix using cloud for global streaming
- Spotify scaling music services worldwide
- Zoom Communications managing remote meetings

Understanding Data packets

Data Packet - OSI Layers

- Each layers wraps the original data in its own Header and Trailer
- It is called encapsulation

Data Packet

- Application Layer # 7
- We have raw data of the user e.g. an email, a web page request
- → data / message

Data Packet

- Presentation Layer # 6
- Data is translated, encrypted and compressed here so the receiving device can interpret it correctly
- → Data / message

Data Packet

- Session layer # 5
- It starts, maintains and terminates the communication connection / sessions between systems
- → data / message

Data Packet

- Transport layer # 4
 - Data from the upper layers is broken down into smaller, manageable chunks
 - It adds port number
 - → Segment (TCP) and datagram (UDP)
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- A port number is a 16-bit logical identifier used to direct network traffic to the correct application or service on a device
 - IP address routes data to the device itself, the port number ensures the data reaches the correct program

Data Packet

- Network layer #3
- Handles routing and logical addressing
- Adds a Source and Destination IP Address
- Routers use this information to determine the best path for the data
- → Packet (in TCP) (or datagram in UDP)

Data Packet

- Data Link Layer # 2
 - This layer is responsible for node-to-node delivery
 - It wraps the packet with Source and Destination MAC Addresses
 - It also adds an error-checking Trailer (Frame Check Sequence) to ensure the data wasn't corrupted in transit
 - → Frame
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- MAC address identifies the physical hardware while an IP address identifies the logical network address of a device. MAC spoofing is possible - mostly temporary or done on random basis

Data Packet

- Physical Layer # 1
- Converts the 1s and 0s of the data Frame into physical electrical, optical, or radio signals
- This layer moves the raw bits over the physical medium (cables or Wi-Fi)
- → Bits